HRLE-SARDet: A Lightweight SAR Target Detection Algorithm Based on Hybrid Representation Learning Enhancement

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***Abstract*— In recent years, deep learning has been widely used in remote sensing, especially in the field of synthetic aperture radar (SAR) image target detection. However, all of these deep learning models continue increasing the network's depth and width without maintaining a good balance between accuracy and speed. Therefore, in this article, we propose a hybrid representation learning-enhanced SAR target detection algorithm based on the unique features of SAR images from a lightweight perspective called HRLE-SARDet. First, we design a lightweight and scattering feature extraction backbone that is more suitable for SAR image data. Second, for the multiscale feature discrepancy, we design a new multiscale feature fusion neck. Next, to better extract the scattering information from small targets of SAR images and improve the detection accuracy, we design a lightweight hybrid representation learning enhancement module. Finally, to better fit target detection for SAR image datasets, we redesign a more flexible loss function, which allows for an easy adjustment of the importance of polynomial bases according to the target task and dataset. Extensive experimental results on three SAR image ship target datasets (SSDD, AIR-SARShip-2.0, and HRSID) and a newly released large multiclass target SAR dataset (MSAR-1.0) show that our HRLE-SARDet achieves 98.4%, 79.2%, 92.5%, and 88.4% mean average precision (mAP) with only 1.09 M parameters and 2.5 G floating-point operations (FLOPs) on the SSDD, AIR-SARShip-2.0, HRSID, and MSAR-1.0 datasets, respectively, which is an excellent performance.**

***Index Terms*— Deep learning, hybrid representation learning enhancement, lightweight, synthetic aperture radar (SAR) target detection.**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging remote sensing device that can capture the Earth's surface around the clock and in all weather conditions. SAR uses techniques, such as virtual arrays and pulse compression, to obtain high-resolution 2-D images of features, which can be used in areas, including target detection and identification. SAR images are widely used for military surveillance, environmental and natural disaster monitoring, sea-surface oil spill detection, maritime transport monitoring, and sea-surface ship detection due to the unique imaging mechanism of the system and its rich electromagnetic scattering characteristics [1], [2], [3]. Among them, researchers' ship detection in SAR images with complex oceanic backgrounds has received increasing attention. In recent years, SAR satellites have collected a large amount of SAR image data, and various organizations have successfully made public SAR ship target detection datasets for researchers to study. Furthermore, due to the rapid development of deep learning, many researchers have increasingly started applying convolutional neural networks (CNNs) in the SAR image ship detection field and achieved good results. However, these convolutional networks are becoming deeper, and they prioritize accuracy but ignore detection's real timeliness, speed, and portability. Therefore, it is necessary to design a lightweight model that can balance speed and accuracy for real-time and fast detection of SAR image ship targets and can be easily deployed to embedded platforms.

Before deep learning gained attention in the field of target detection on SAR images, researchers used traditional methods to perform target detection of SAR images. For example, SAR image contrast information-based [4], [5], [6], SAR image geometric and texture feature-based [7], [8], and SAR image pixel statistical analysis-based methods [9], [10] are the early standard traditional SAR target detection methods. Later, Leng et al. [11] proposed the constant false alarm rate (CFAR) algorithm since they could adaptively determine the detection threshold based on the given false alarm rate and statistical distribution of background clutter. This algorithm is also the most traditional detection method in the field of target detection on SAR images. Researchers have continued to innovate and upgrade based on their algorithm until today. For example,

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Liu et al. [12] proposed a sea–land separation method to automatically select the detection area, which greatly reduces the false alarm rate and improves the detection accuracy. The performance of these traditional methods is largely influenced by the statistical modeling of sea clutter and the parameter estimation of the selected model. The computational process becomes cumbersome with increasing model complexity and poor migration capability.

Due to the rapid development of deep learning algorithms in recent years, many researchers have started to combine deep learning algorithms with SAR image target detection and achieved a series of amazing results [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. These deep learning target detection algorithms are diverse with different functions and application scenarios. Nevertheless, they can be broadly classified into two categories based on the presence or absence of proposal boxes: 1) single-stage detection algorithms without proposal boxes and 2) two-stage algorithms based on proposal boxes. In general, compared with single-stage target detectors, two-stage target detectors can achieve higher accuracy, but the training and detection speed is often less than those of single-stage algorithms. Many SAR ship target detectors increasingly choose single-stage algorithms to ensure real-time detection and recognition. However, it has also been shown that single-stage target detection methods are more prone to false detections and localization errors when detecting small targets. Therefore, to consider real-time detection in practical engineering applications, we must also consider solving the poor detection of small targets in a single stage.

Although these target detection algorithms are more rationally designed, the networks are designed even deeper, increasing the target detection accuracy. The algorithms are specifically designed for optical images, and these target detectors are designed to improve the accuracy. Therefore, there are very few detectors designed explicitly for SAR images, and if we apply these target detection algorithms for optical images directly to SAR images, the results may not be as good as expected because: 1) SAR images have a more complex imaging mechanism than optical images, and the targets are sparser and consist of many strong scattering points, which do not have the same rich imaging information as optical images; 2) the target size of SAR images is more different than that of optical images, and because the imaging angle of SAR images is larger than that of optical images, the imaging span of SAR images is larger; therefore, the ship targets in many SAR images are only a dozen pixels in size, showing only one or two strong scattering points; and 3) since SAR images are shown in the form of scattering points, coupled with their complex land background, ship targets at sea are easily disturbed by land background clutter, and ships docked on the shore are more likely to blend in with the background, which results in missed detection.

According to the above introduction, it is necessary and valuable to design a lightweight single-stage target detection algorithm for SAR image target detection that is easy to deploy on embedded platforms. This article innovatively proposes various lightweight SAR target detection algorithms based on hybrid representation learning enhancement, HRLE-SARDet, which solves the SAR image ship-target detection problem from a more balanced perspective and improves detection accuracy while substantially reducing the number of parameters and floating-point operations (FLOPs). The main contributions of this article are summarized as follows.

1) To develop a more suitable SAR image target detection algorithm for embedded platforms, we design LSFEBackbone, a more reasonable lightweight and feature extraction for SAR image datasets to better fit the characteristics of SAR image ship data. Its number of parameters and FLOPs to maintain good performance is greatly reduced compared to CSPDarknet53.
2) For the characteristics of multiscale and large-scale variation of SAR ship targets, we redesign its multiscale neck, BiFPF, to better fully integrate the feature information of different scales and improve the detection effect of ship targets of different scales.
3) To better extract the scattering information of small targets and improve their detection accuracy, we design a hybrid representation learning enhancement module called HRLE-C3, which combines the respective advantages of a CNN and self-attention to complement each other. As a result, the design can improve the detection performance of small SAR targets while reducing the number of parameters and FLOPs.
4) For a more suitable target detection task on the SAR image dataset, we redesign a more flexible loss function, PBCE, based on the BCE loss function. We regard and design it as a linear combination of polynomial functions to approximate the function by the Taylor expansion. As a result, PBCE allows the importance of the polynomial basis to be easily adjusted according to the target task and dataset, which makes our model better suited for training on SAR images.

II. Related Work

Because the existing public SAR target detection datasets are SAR-ship target detection datasets, such as SSDD [23], AIR-SARShip-2.0 [24], and HRSID [25], target detection on SAR images is generally performed for ship target detection. The existing traditional SAR ship target detection methods were developed based on CFAR [11], which can adaptively determine the detection threshold based on a given false alarm rate and the statistical distribution of background clutter. Later, Wang and Liu [26] proposed an intensity-space domain CFAR algorithm by integrating intensity and spatial information on its basis. However, SAR images often have very complex backgrounds due to their imaging specificity; for instance, ship targets may be in a complex ocean background and dock in a complex land background. As a result, these algorithms often cannot handle these scenes well, and complex background clutter statistics often interfere with the algorithm's performance. Thus, Huo et al. [27] selected candidate detection regions by calculating exceptionally stable extreme value regions and obtained thresholds to detect ships. Similarly, Shi et al. [28] separated ships from the background by calculating the directed gradient histogram, which led to better feature extraction. However, these traditional methods

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are often ineffective in detecting small targets, require manual parameter setting and adjustment, and heavily rely on manual design, which is highly undesirable. As a result, the robustness of these traditional algorithms is very poor, and they cannot be applied well to migrate to other SAR datasets. Furthermore, the detection speed and real-time performance also cannot satisfy the requirements of engineering applications.

Therefore, the end-to-end advantages of deep learning-based algorithms are becoming increasingly obvious. In recent years, with the continuous updating of deep learning technology, many researchers have chosen to use deep learning algorithms to perform SAR image target detection. In contrast to traditional hand-designed feature-based SAR image target detection algorithms, deep learning algorithms do not need to artificially design complex feature extraction processes. Therefore, their training and detection processes are simpler and more convenient. They only need to consider how to design a robust network structure to extract the features of SAR targets well. Thus, increasing attention is focused on using CNNs to handle SAR image targets, especially popular CNN methods for SAR image ship targets. For example, to address the multiscale problem of SAR image ships, Liu et al. [29] constructed a ship proposal box generator to improve the recall rate and proposal box quality. Kang et al. [13] applied a multilayer fusion-based contextual region CNN in the field of SAR ship detection to solve the problem of small target misdetection of SAR ship images and improve detection performance. By proposing an attention-guided balanced pyramid structure and fusing it with a refined detection head, Fu et al. [30] balanced deep and shallow semantic features at different scales, allowing the detection head to focus more effectively on small boats in complex scenes. However, these detection methods often rely on a very deep network model structure and do not consider practical engineering applications; the number of parameters, FLOPs, and detection accuracy are not well-balanced.

Therefore, many people have recently started to work on lightweight SAR ship target detection models. Chang et al. [31] introduced YOLOv2 and improved it to design an SAR ship detector with fewer parameters. Although the detection speed is improved, the theoretical explanation is vague, and the performance of the YOLOv2 model is not very good. Zhou et al. [32] designed a lightweight meta-learning method for few-shot object detection on SAR images based on YOLOv3 [33]. Recently, Wan et al. [20] proposed a lightweight anchor-free frame SAR ship detection method that reduced the number of parameters of the model. Similarly, Feng et al. [22] combined position-enhanced attention with the latest target detector YOLOX [34] and performed a lightweight design on the backbone, which reduced the number of parameters and FLOPs and ensured good performance. Although these methods have contributed to the design of lightweight SAR ship detectors, they have not truly achieved a proper balance between the accuracy and lightness of the model. Moreover, they do not consider the need to design detection models for SAR image target detection datasets.

Therefore, considering the needs of real engineering applications, we innovatively design a lightweight and high-precision SAR target detection algorithm for SAR image target detection datasets, which is suitable to transplant to embedded platforms.

III. Methods

This article proposes a lightweight SAR target detection algorithm based on hybrid representation learning enhancement called HRLE-SARDet. As a result, the most balanced level in terms of accuracy and lightweight degree is achieved. The HRLE-SARDet model mainly includes the following modules and strategies: the state-of-the-art (SOTA) target detection benchmark framework, YOLOV5 [35], the more reasonably designed SAR image scattering feature extraction lightweight backbone LSFEBackbone, the multiscale deep and shallow semantic feature fusion neck, BiFPF, the hybrid representation learning enhancement module, HRLE-C3, and the more flexible loss function, PBCE.

A. Overall Network Architecture of HRLE-SARDet

The overall network architecture of our proposed HRLE-SARDet model is shown in Fig. 1. The model consists of five parts in the figure: input, backbone, neck, head, and output. Specifically, the input image first enters our designed lightweight backbone network, LSFEBackbone, which can more reasonably extract the discrete scattering features of SAR images. Then, the feature maps extracted from the backbone enter our designed complex multidirectional multiscale scattering feature fusion Neck-BiFPF, which fully fuses deep and shallow semantic features at different scales to better learn the information of targets at various scales. BiFPF learns the weights of different resolution input features. Then, we introduce the representation learning enhancement module, HRLE-C3, before extracting the output of small target-scale features, which combines the respective advantages of CNN and self-attention to complement each other and can improve the detection performance of SAR small targets while reducing the number of parameters and FLOPs. The final output of the three scales of the predicted feature maps is subjected to loss calculation by our PBCE loss designed for SAR image training learning to achieve the most suitable training state. We introduce the improvements of these four aspects in more detail later.

B. Lightweight Scattering Feature Extraction Backbone: LSFEBackbone

Most of the existing backbone structure is designed to be deeper, and the number of parameters and FLOPs of the network increase, which often causes computational resources to be wasted and overfitting when encountering models with small datasets. It is not particularly suitable for deployment to mobile devices. It is usually designed to extract the features of optical images, which is not necessarily suitable for directly extracting the discrete scattering feature information of SAR images. Since SAR datasets are usually sparse and consider the characteristics of their own targets, we specifically design

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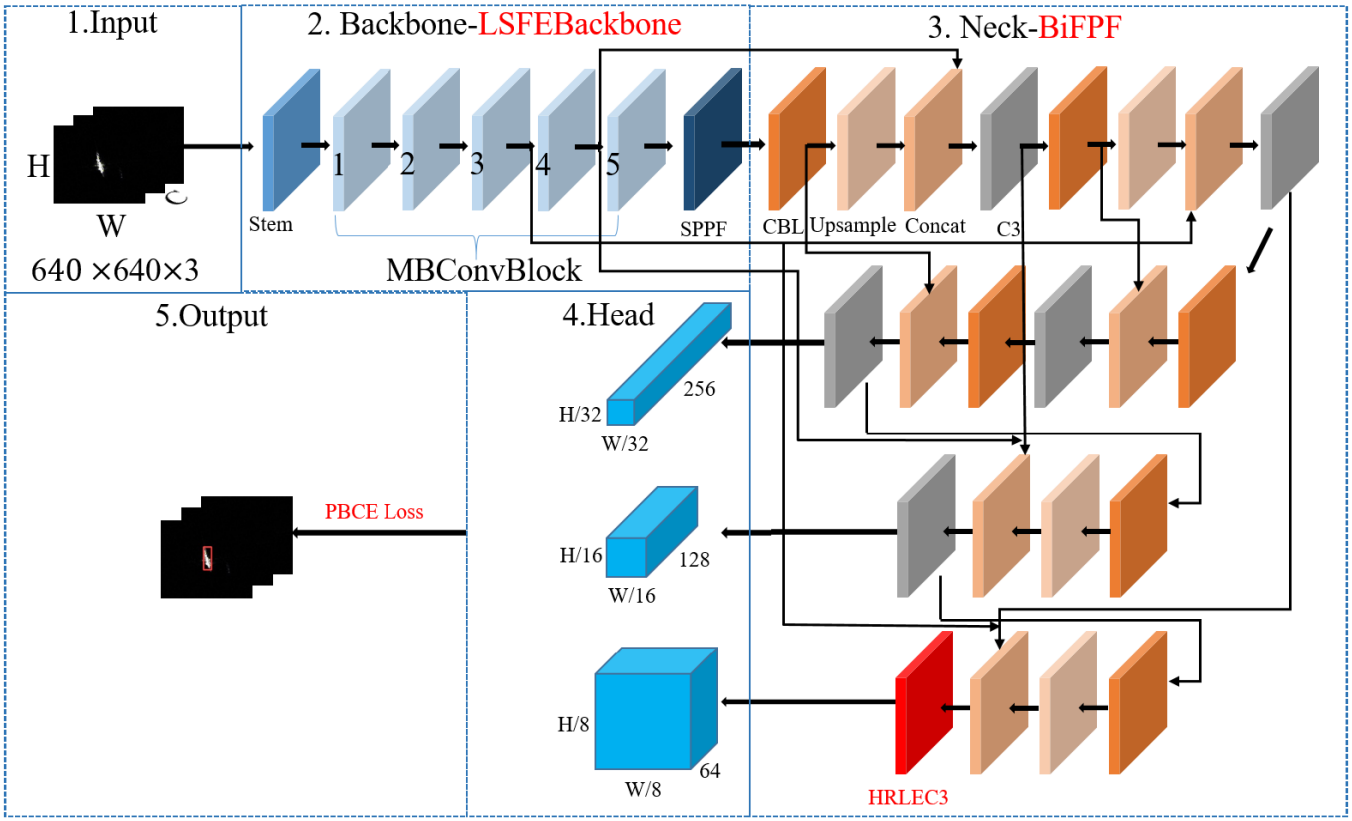

Fig. 1. Overall network architecture of the HRLE-SARDet model.

a new lightweight scattering feature extraction backbone network called LSFEBackbone, which has fewer parameters and FLOPs.

LSFEBackbone is an improved version of EfficientNet-Lite [36], the end-side version of EfficientNet released by Google. Indeed, the accuracy, latency, and parameters of EfficientNet-lite have left MobileNet V2 [37], ResNet 50 [38], Inception v4 [39], and other models far behind.

We introduce the smallest backbone version of EfficientNet-Lite, i.e., EfficientNet-Lite0, and analyze and improve it with extensive experimental validation to design a lightweight backbone, LSEBackbone, which is more suitable for extracting scattering features from SAR datasets. The main aspects are given as follows.

1) We continue to use MBConvBlock of EfficientNet-Lite0 and redesign and stack it to remove unnecessary MBConvBlocks.
2) To increase the convolutional kernel to extract features suitable for the sensory field, we use all large 5 × 5 convolutional kernels to extract features after the end of stage 2 of the network.
3) We use the same improved lightweight squeeze-and-excitation (SE) [40] version as MoileNetV3 [41] and change its sigmoid activation function to the less computationally intensive H-sigmoid activation function as follows:

$$H\text{-sigmoid} = \frac{\text{ReLU6}(x+3)}{6}. \tag{1}$$

4) Simultaneously, the SE module is activated by the rectified linear unit (ReLU) unsaturated activation function after the first 1 × 1 convolution, which has certain defects. The ReLU has a gradient of 0 for $x < 0$, and this neuron may never be activated by any data again. Meanwhile, in a parametric rectified linear unit (PReLU) [42], the slope of the negative part is based on the data and is not predefined. Therefore, it only adds a few parameters and FLOPs, so the risk of network overfitting only slightly increases. In particular, when the same a_i is used for different channels, there are even fewer parameters. Therefore, we replace the ReLU activation function in SE and the ReLU6 activation function in MBConvBlock with PReLU, as shown in Fig. 2. We call the improved SE the PHSE in Fig. 3 and verify that it works after extensive experiments.

The structural details of LSFEBackbone are shown in Fig. 4, and the specific parameters are shown in Table I. The results of extensive experimental validation employing the SAR dataset demonstrate our designed backbone's strong feature extractability.

C. Multiscale Scattering Feature Fusion Neck: BiFPF

The third part in Fig. 1 shows our multiscale scattering feature fusion neck, BiFPF, designed explicitly for SAR image datasets. BiFPF can sufficiently fuse deep and shallow semantic features at different scales to better learn information from targets at different scales. It adds various resolution

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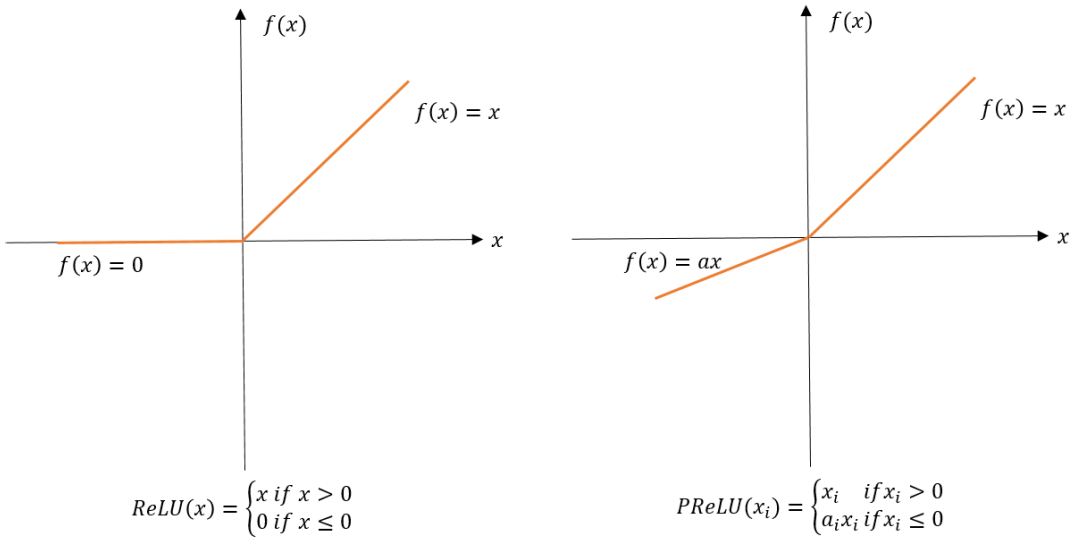

Fig. 2. Comparison of two activation functions. (Left) Schematic of ReLU activation function. (Right) Schematic of PReLU activation function.

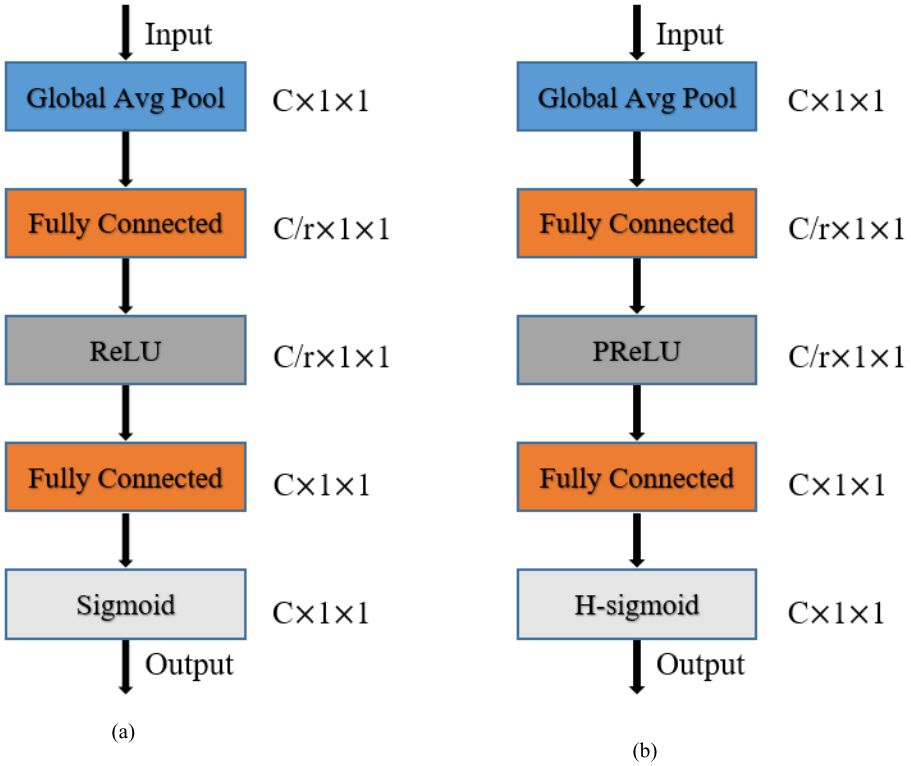

Fig. 3. Comparison of the structure of (a) original SE module and (b) modified PHSE module.

inputs and learns the weights of different resolution input features. It is more challenging to detect small-scale ships in SAR image target detection because, in the convolution downsampling process, there are more scattered pixel points of large ships and fewer scattered pixel points of small ships. When the convolution becomes deeper, the features of large ships are easily retained, and the features of small ships are more easily ignored. Therefore, to enhance the target detection performance and feature extraction for small-scale targets, we design a characterization learning enhancement module, HRLE-C3, and apply it before the output of small-scale feature prediction maps, which combines the respective advantages of CNN and self-attention to complement each other and can focus on extracting the features of SAR small-scale targets while reducing the number of parameters and FLOPs and improving the detection accuracy of small targets of SAR images. In addition, the designed multiscale scattering feature fusion neck module BiFPF can better fuse the feature

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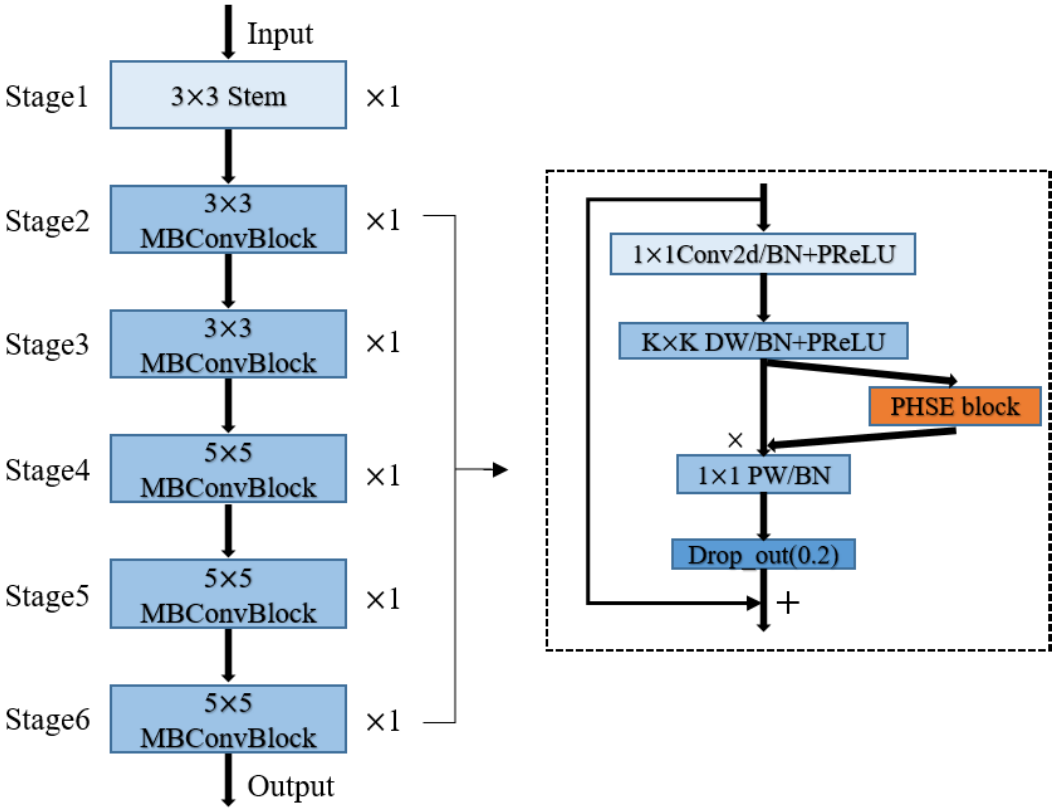

Fig. 4. Details of the LSFEBackbone network structure.

TABLE I
PARAMETERS OF LSFEBACKBONE NETWORKS STRUCTURE

	Out Channel	Kernel Size	Stride	expand	Dropout rate	PHSE
Stem	32	3	2	-	-	-
MBConvBlock	16	3	1	6	0.2	-
MBConvBlock	24	3	2	6	0.2	-
MBConvBlock	24	3	1	6	0.2	-
MBConvBlock	40	3	2	6	0.2	-
MBConvBlock	40	3	1	6	0.2	-
MBConvBlock	80	5	2	6	0.2	√
MBConvBlock	80	5	1	6	0.2	√
MBConvBlock	112	5	1	6	0.2	√
MBConvBlock	192	5	2	6	0.2	√
MBConvBlock	320	5	1	6	0.2	√
SPPF	1024	5	1	-	-	-

map information of different scales, which is more suitable for extracting features of small-scale ship targets. The basic structure of the BiFPF design is shown in Fig. 5.

The structures of BiFPF and bidirectional feature pyramid network (BiFPN) proposed recently by Tan et al. [43] are somewhat similar but still different. BiFPN is improved on the original PANet [44]. As shown in Fig. 6, we can see obvious differences between our BiFPF and BiFPN. First, BiFPN is designed for experimental testing and adjustment on optical datasets, while our BiFPF is designed for experimental testing and adjustment on SAR image datasets. Second, the output feature fusion stage of BiFPN involves the seventh stage, while our model is designed from a lightweight perspective, and there are a few stages of the feature output fusion layer. Finally, our BiFPF adds an upsampling structure of the feature pyramid network (FPN) [45] and adds a feature learning

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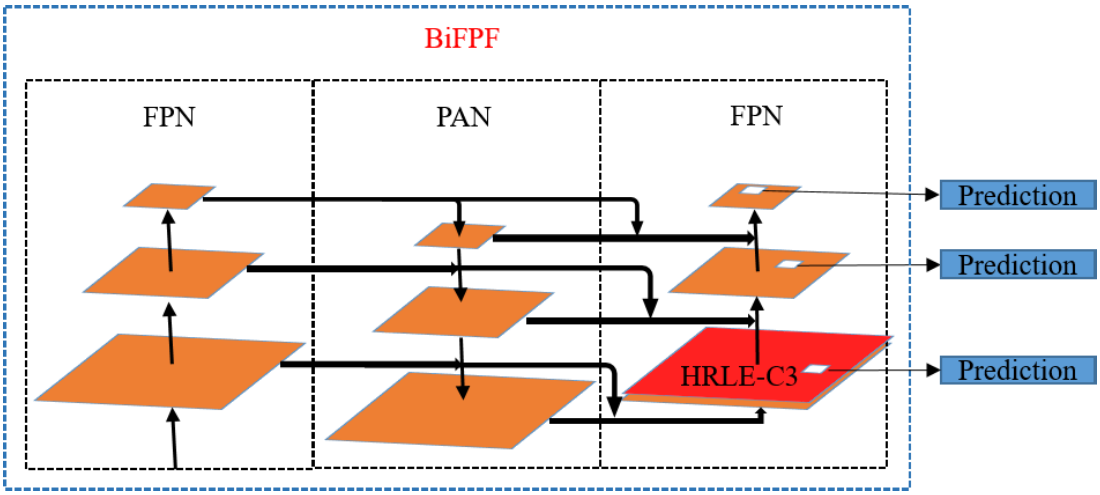

Fig. 5. Basic structure of BiFPF.

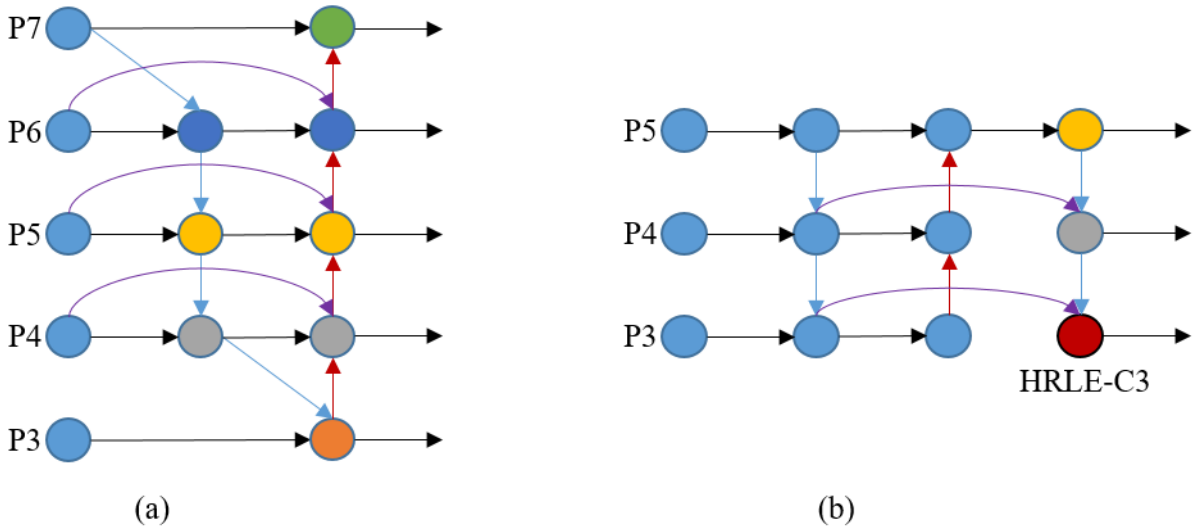

Fig. 6. Comparison between (a) BiFPN and (b) our BiFPF.

enhancement module, HRLE-C3, before the output of the small-scale feature prediction map in the P3 stage, which can focus on extracting the features of SAR small targets, reduce the number of parameters and FLOPs, and improve the detection accuracy of small targets in SAR images.

D. Hybrid Representation Learning Enhancement Module: HRLE-C3

In recent years, CNNs and self-attention networks have been the hottest research hotspots in the field of computer vision (CV). In CV fields, such as image classification, target detection, and target tracking, CNNs have reached impressive achievements and realized the SOTA performance on various datasets. Simultaneously, with the explosion of vision transformers [46], transformers based on various forms of self-attention, began to show excellent performance across multiple vision tasks, and many more exceeded the performance of the CNNs.

At present, due to different design theories, convolution and self-attention are often considered two different methods, but both are powerful representational learning methods in the field of deep learning. By detailing the decomposition process of these two modules, Pan et al. [47] recently showed that they had remarkable similarities in the 1×1 convolution operation. It revealed a very close connection between convolution and self-attention, i.e., convolution and self-attention are practically identical in generating a feature map via 1×1 convolutions. Based on the results of this study, a hybrid learning module called ACmix was developed and integrated convolution and self-attention with minimal computational overhead. ACmix can combine the advantages of both modules while avoiding two expensive mapping operations. Fig. 7 shows the schematic of the ACmix structure, which contains two stages. Stage I is the convolution path to obtain a rich set of intermediate features containing $3 \times N$ feature mappings, whose inputs are mapped by three 1×1 convolutions on the input features and reshaped to N pieces. Stage II is a

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self-attention path with a collection of N sets of intermediate features, with a total of three feature maps from a 1×1 convolution in each set. As in the traditional multiheaded self-attentive module, query, key, and value correspond to the three 1×1 convolution feature maps. The fully connected layer is used to generate N feature maps to process the convolutional paths with convolutional kernel size k. Thus, it can process the input features and the shifted-and-aggregated generated features with convolution so that information can be collected from the local perceptual field as in the traditional one.

Finally, the outputs of the two stages are summed, and the two learnable scalar controls can be their intensities as follows:

$$F_{\text{out}} = \alpha F_{\text{conv}} + \beta F_{\text{self-att}}. \tag{2}$$

Since ACmix combines the respective advantages of convolution and self-attention, it is suitable for application in SAR image target detection for hybrid representation learning enhancement. Thus, we refer ACmix to our model and modify its original C3 module of YOLOV5 to design the hybrid representation learning enhancement module HRLE-C3, as shown in Fig. 8, considering the problems of small targets of SAR images being difficult to detect and other problems. It is inserted into the output in Figs. 5 or 6 before the large-size feature map that can easily detect small targets, and it is combined with the BiFPF module to enhance the feature extractability and detectability of small targets.

E. Poly BCELoss: PBCE

Both classification loss and object loss use the most common binary cross-entropy loss (BCELoss) function. However, it does not work well when applied directly to SAR image datasets. Therefore, it is valuable to design suitable loss functions specifically for the target detection task of SAR image datasets. Recently, Leng et al. [48] viewed and designed the loss function as a linear combination of polynomial functions, approximated the function by Taylor expansions, and proposed a new and simple loss function called PolyLoss. Therefore, we use the idea of PolyLoss to modify BCELoss and obtain the loss function, PBCE, which can easily adjust the importance of polynomial bases according to SAR image datasets and the target detection task.

BCEloss often struggles with overfitting problems and cannot always be applied to other tasks. In recent years, new loss functions have been explored by many researchers, mostly using meta-learning, integration, or synthesis of different losses. Leng believes that the commonly used classification loss and object loss function can be decomposed into a series of weighted polynomial bases in the following form:

$$\text{Loss} = \sum_{i}^{\infty} \alpha_i (1 - P_t)^j \tag{3}$$

where α_i is the positive coefficient in the real domain of the polynomial, P_t is the predicted probability of the target class, and each polynomial basis $(1 - P_t)^j$ is weighted by the corresponding polynomial coefficient α_i. The latter coefficient allows the polynomial-based loss function to easily adjust its polynomial basis according to its polynomial coefficients. The polynomial coefficients α_i must be adjusted if better results are desired on different datasets and tasks. Finally, the experiment shows that only one polynomial coefficient must be adjusted to achieve better performance than the ordinary loss function. Therefore, our PBCE uses its ideas for reference and starts from the original BCE to transform it. The formula of BCE is given as follows:

$$L_{\text{BCE}} = \begin{cases} -\log\ p_t, & \text{if } y = 1 \\ -\log(1 - p_t), & \text{otherwise.} \end{cases} \tag{4}$$

From the above idea and Taylor series expansion, the BCE loss function is observed in the form of a polynomial series and can be expanded as follows:

$$L_{\text{BCE}} = \sum_{i=1}^{\infty} \alpha_i (1 - p_t)^i. \tag{5}$$

According to the mathematical knowledge of the Taylor series, its polynomial coefficient α_i is $(1/i)$, so the BCE loss function can be expressed as follows:

$$L_{\text{BCE}} = \sum_{i=1}^{\infty} \frac{1}{i}(1 - p_t)^i = (1 - p_t) + \frac{1}{2}(1 - p_t)^2 + \cdots \tag{6}$$

We use the gradient descent method to optimize the BCE loss, so we must derive and expand p_t to obtain the following:

$$-\frac{\partial L_{\text{BCE}}}{\partial p_t} = \sum_{i=1}^{\infty}(1 - p_t)^{i-1} = 1 + (1 - p_t) + (1 - p_t)^2 + \cdots \tag{7}$$

From its derived expansion polynomial, we know that it has a different sensitivity to p_t. The first polynomial is a constant term with a value of 1, which provides a constant gradient for BCE loss and is independent of the size of p_t. When the training ground becomes deeper, i.e., $i \gg 1$, the predicted value of p_t is infinitely close to 1, so the later polynomial approaches 0 and is strongly suppressed.

In general, the Taylor series expansion of a loss has infinitely many polynomial coefficients, but it is impossible for us to adjust all of them. Therefore, it is not feasible to optimize the coefficients of the entire loss, so we can achieve the desired effect by adjusting only the first polynomial term as in Leng et al. [48] Therefore, the PBCE can be expressed as follows:

$$L_{\text{PBCE}} = (1 + \alpha)(1 - p_t) + \frac{1}{2}(1 - p_t)^2 + \cdots = L_{\text{BCE}} + \alpha(1 - p_t). \tag{8}$$

In the experiment, we take the value of α as 1. Finally, the experimental results show that our PBCE loss design is effective.

IV. Experiments

In the experiment, to validate the performance of our HRLE-SARDet, we performed extensive ablation experiments on three public benchmark SAR ship target detection datasets (SSDD [23], AIR-SARShip-2.0 [24], and HRSID

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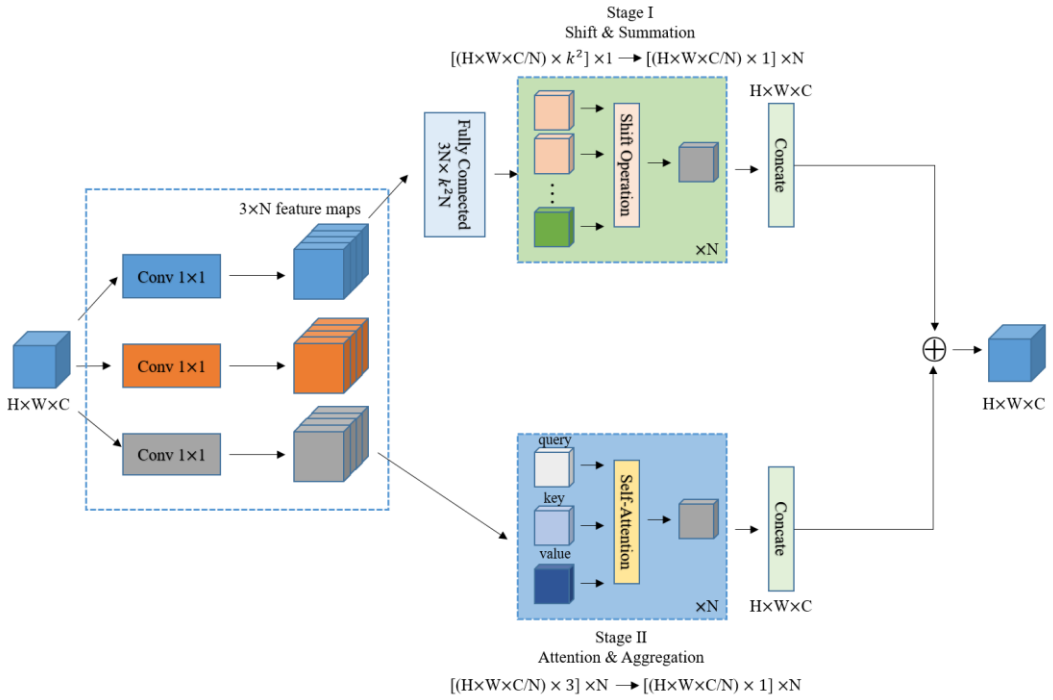

Fig. 7. Structure of ACmix.

[25]) and a newly released large multiclass target SAR dataset (MSAR-1.0) [21], [49] for validation. Furthermore, we compared our proposed HRLE-SARDet with the existing SOTA detection methods, and the performance of HRLE-SARDet proved to be superior.

A. Dataset Introduction and Experimental Settings

To most accurately evaluate the effectiveness and performance of our HRLE-SARDet model, we conducted extensive validation experiments on three commonly used SAR image ship target detection datasets (SSDD, AIR-SARShip-2.0, and HRSID) and a recently released large-scale multiclass SAR target detection dataset (MSAR-1.0).

1) SSDD: The SSDD dataset is a PASCAL visual object classes (PASCAL VOCs) annotated format dataset, so the code using the PASCAL VOC annotated format dataset can directly train SSDD. The ships in SSDD can be classified into large, medium, and small targets according to the target size, and small targets make up most of the SSDD dataset. SSDD contains the imaging results from TerraSAR-X, RadarSat-2, and Sentinel-1 under various satellite sensors with four different polarizations (HH, VV, HV, and VH), and the resolution is 1–15 m. The richness of the data enables the detector to be trained for more robust performance, but it also makes detection more difficult, and the detection performance decreases. The SSDD dataset contains 1160 images with 2456 ships, so on average, each image contains 2.12 ships. In this article, according to strict regulations, images with the last digits of file numbers 1 and 9 are identified as the test set. The remainder is considered a training set, dividing the dataset into a ratio close to 8:2. Thus, there are 928 images in the training set and 232 images in the test set. This maintains the same data distribution as the other methods and facilitates a fair comparison.

2) AIR-SARShip-2.0: High-resolution SAR ship objection detection dataset-2.0 (AIR-SARShip-2.0) releases 300 images. The resolution of the image includes 1 and 3 m, the imaging mode includes bunching and striping, the polarization mode is single polarization, and the polarization mode is VV. The scene type includes ports, islands and reefs, and the sea surface with different levels of sea conditions, and the target covers more than ten categories of thousands of ships, such as transport ships, oil tankers, and fishing boats.

In the dataset, each image size is 1000 × 1000 pixels, and the image format adopts the Tiff format, single channel, and 16-bit image depth. The corresponding annotation file provides details, such as the length and width of the corresponding image, the category of the annotation target, and the location of the annotation rectangle.

Because the bit depth of the image is 16 bits and the image is a nearly black scene, which is difficult to recognize with the naked eye and model training, we need to preprocess it and convert its bit depth from 16 to 8 bits. In addition, the dataset was not divided into training and verification proportions when it was released, so we divided it into a training set and verification set according to the proportion 8:2, that is, the training set contains 240 pictures and the verification set contains 60 pictures.

3) HRSID: To validate the ship detectability of our proposed method under different datasets, we also validated the performance of our method on another large-scale high-resolution SAR image ship target detection dataset, i.e., the HRSID dataset, which enables tasks, such as target detection and instance segmentation. Panoramic images with a

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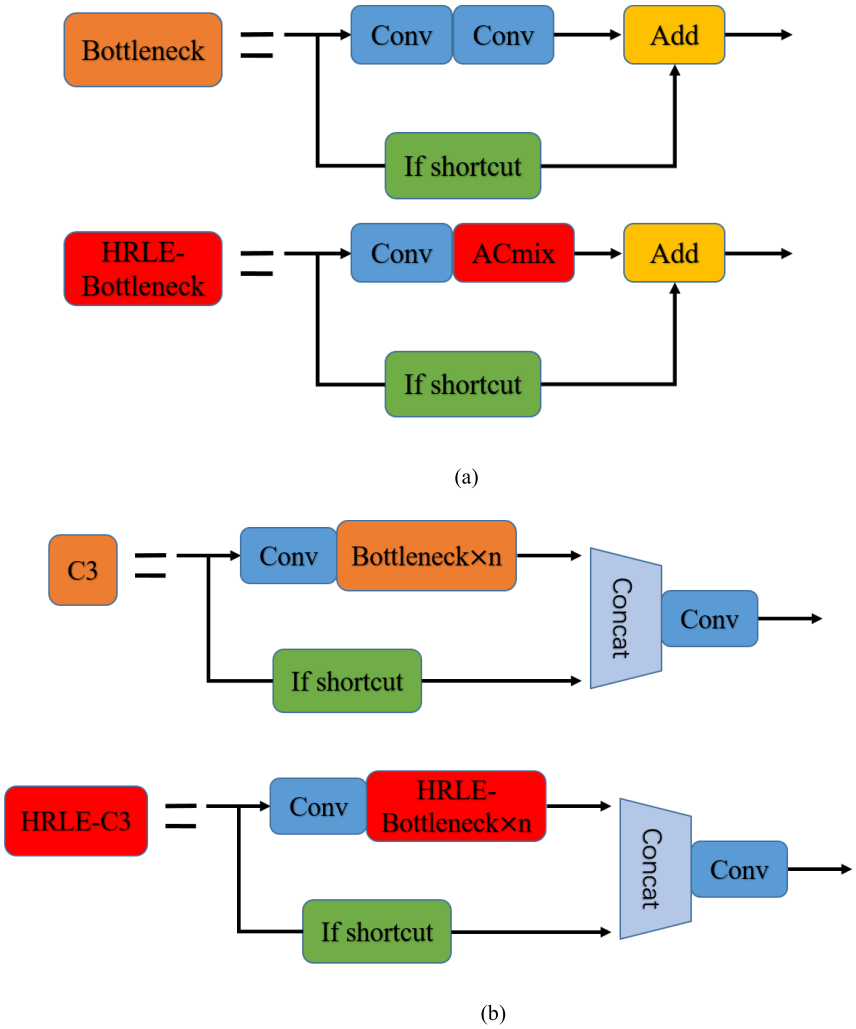

Fig. 8. Comparison of HRLE-C3 of HRLE-SARDet and the C3 structure of YOLOV5. (a) Bottleneck of YOLOV5 and the HRLE-Bottleneck structure of HRLE-SARDet. (b) C3 of YOLOV5 and the HRLE-C3 structure of HRLE-SARDet.

resolution of 1–5 m were cropped to a size of 800 × 800. In total, 5604 cropped SAR images containing 16 951 ships were divided by HRSID in a ratio of 6.5:3.5 between the training and test sets.

4) MSAR-1.0: In addition, to verify our model's generalized classification detectability under the multiclass SAR dataset, we introduced a newly released large multiclass target SAR dataset MSAR-1.0 [21], [49] for experiments, which is a standard SAR target dataset constructed from the data collected by the HISEA-1 and Gaofen-3 satellites. It includes 28 449 images containing multiple polarization methods and multiple scenes, with a total of four types of targets: ship, oil tank, airplane, and bridge. Since the training and validation ratios were not specified in the original paper [21], we redivided them into training and validation sets at a ratio of 8:2 and rechecked some mislabeling. The divided experimental dataset for experimental comparison will be released at a later time.

The model has basically identical training parameters to YOLOV5, the batch size is set to 16, the image size of SSDD training is set to 640 × 640, the image size of the AIR-SARShip-2.0 Dataset training is set to 1024 × 1024, the image size of the HRSID training is set to 800 × 800, the image size of the MSAR-1.0 for training is consistent with its published paper [21] at 640 × 640, and the remaining comparison experiments are consistent. This experiment was conducted with a NVIDIA GeForce RTX 2060 super, Windows 10 system, Python version 3.7, PyTorch version 1.7.1, and CUDA version 10.1. Fig. 9 shows the selected examples from four SAR image datasets. The basic information of the experimental data and experimental settings are shown in Table II.

B. Experimental Index

The evaluation index criteria in this experiment are visual object classes (VOCs) evaluation criteria. Thus, to evaluate the detection performance of our proposed method, we use the average accuracy mAP50 index. The mean average precision (mAP) is calculated from the accuracy P and recall R, where accuracy is the percentage of correct predictions among the samples that are predicted as positive examples. This percentage is calculated from true positives (TPs) and false positives (FPs) as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{9}$$

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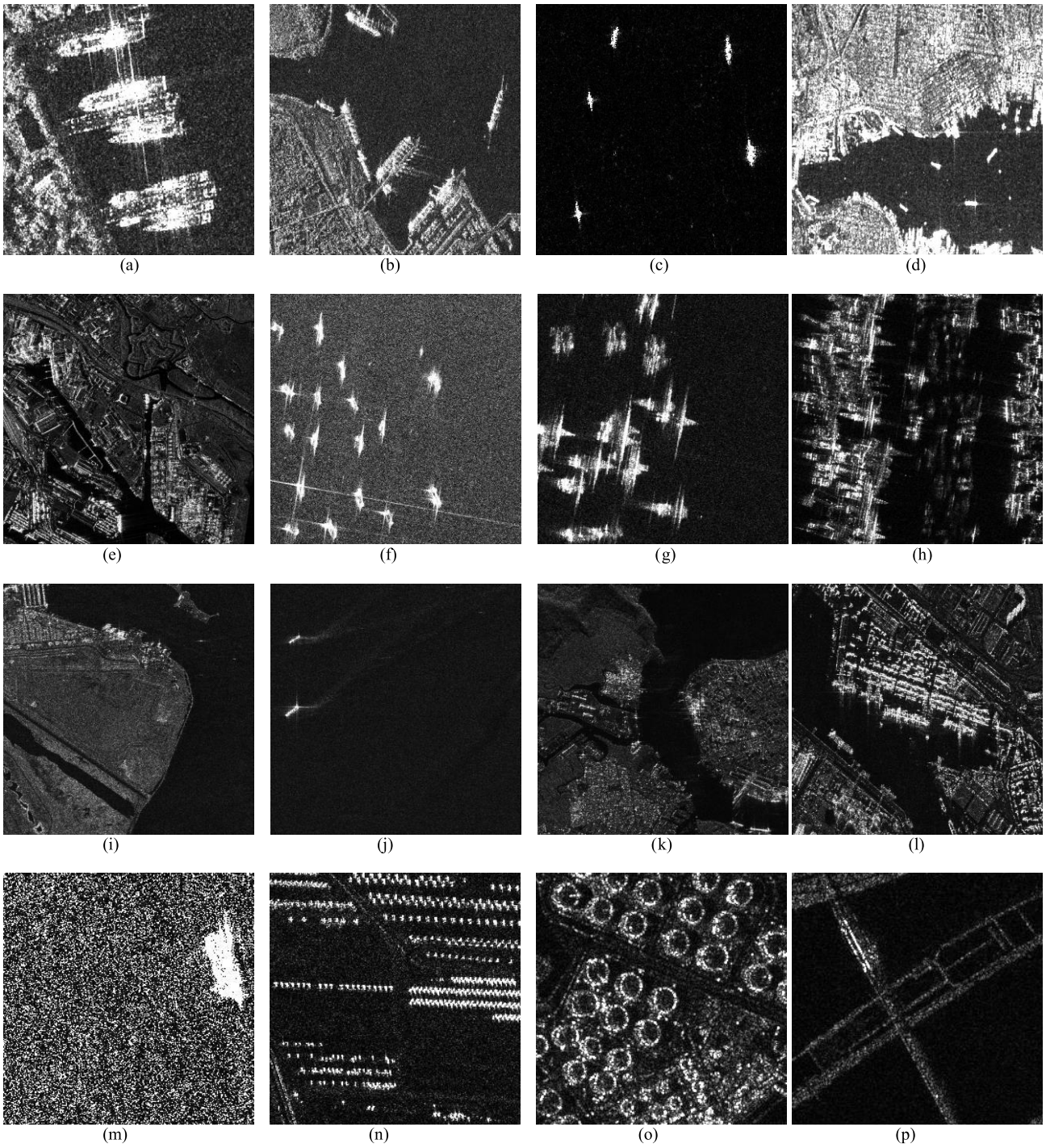

Fig. 9. Selected examples from four SAR image datasets. Some selected examples from (a)–(d) SSDD, (e)–(h) AIR-SARShip-2.0, (i)–(l) HRSID, and (m)–(p) MSAR-1.0.

Correspondingly, the recall rate refers to the probability that the predicted positive examples account for all TP examples in the samples of actual positive examples. The probability is calculated by TP and false negative (FN) as follows:

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{10}$$

Therefore, the value of mAP can be obtained from the area of the PR curve, which is the integral area of PR as follows:

$$\mathrm{mAP} = \int_0^1 P(R) dR. \tag{11}$$

$F1$ is also a common evaluation index. It can also combine precision and recall metrics to comprehensively evaluate the quality of the model

$$F1 = 2 \times \frac{P \times R}{P + R}. \tag{12}$$

In the field of deep learning, when designing a model, it is often necessary to evaluate the parameter size and FLOPs of the model according to the actual application requirements. Therefore, we also calculated the parameter size, FLOPs, and model volume.

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TABLE II
BASIC INFORMATION AND EXPERIMENTAL SETTINGS OF FOUR EXPERIMENTAL DATASETS

	SSDD	AIR-SARShip-2.0	HRSID	MSAR-1.0
images size for training	640×640	1024×1024	800×800	640×640
train: valid	8:2	8:2	6.5:3.5	8:2
number of class	1	1	1	4
class	ship	ship	ship	airplane, bridge, oil-tank, ship
batch size	16	16	16	16
epoch	300	300	600	600

TABLE III
RESULTS OF THE ABLATION EXPERIMENTS ON THE SSDD DATASET

Model	LSFEBackbone	BiFPF	HRLE-C3	PBCE	Params (M)	FLOPs (G)	mAP (%)
YOLOV5n [35]	×	×	×	×	1.76	4.2	97.7
HRLE-SARDet	√	×	×	×	0.94	2.0	97.9
HRLE-SARDet	√	√	×	×	1.10	2.6	98.0
HRLE-SARDet	√	√	√	×	1.09	2.5	98.1
HRLE-SARDet	√	√	√	√	1.09	2.5	98.4

1) Parameter: It is used to describe the size of the model, similar to the spatial complexity of the algorithm. The number of parameters depends on the amount of video memory. It mainly includes all layers of the model, such as the convolution layer, BN layer and fully connected layer, and the total number of weight parameters in the visual network components.

2) FLOPs: This is the abbreviation of floating-point operations, which refers to floating-point operands and is understood as a calculation quantity. It can be used to measure the complexity of the algorithm or model.

C. Experimental Results

To validate the effectiveness and performance of our proposed method, we conducted extensive experimental validation on three SAR image ship target detection datasets (SSDD, AIR-SARShip-2.0 Dataset, and HRSID) and a newly released large multiclass target SAR dataset (MSAR-1.0). In addition, we conducted a series of ablation experiments on SSDD to demonstrate the effectiveness of each of our proposed modules and improved methods. Finally, we compared our proposed HRLE-SARDet with SOTA SAR target detection methods, demonstrating our HRLE-SARDet performance's superiority.

1) Ablation Experiments on the SSDD Dataset: To verify that our proposed modules and the improved method actually work, we conducted a series of ablation experiments on the SSDD dataset for validation. The ablation experiment mainly consists of the following aspects.

1) We replaced the backbone, DSPDarkNet53, of the benchmark, YOLOV5, with our designed lightweight, discrete scattering feature extraction backbone, LSFEBackbone.
2) We replaced the original neck with our designed multiscale scattering feature fusion neck, BiFPF.
3) We replaced the original small target detection head output before the C3 block was replaced with our designed hybrid representation learning enhancement module, HRLE-C3.
4) We replaced the original BCE loss with our PBCE loss designed for SAR image target detection datasets. The experimental results are shown in Table III.

From the results of the ablation experiments, it is clear that the following conditions hold.

1) After replacing LSFEBackbone with our designed lightweight discrete scattering feature extraction backbone, compared with YOLOV5n, the mAP still increased by 0.2% when the FLOPs decreased from 4.2 to 2.0 G, and the number of parameters decreased from 1.76 M to 0.94 M, which proves that our designed lightweight discrete scattering feature extraction backbone, LSFEBackbone, still has stronger feature extractability with fewer parameters and FLOPs.
2) After replacing the original YOLOv5 neck with our designed multiscale scattering feature fusion neck, BiFPF, the mAP increased by 0.1%, although the number of FLOPs and parameters slightly increased compared to that in ablation 1, and we showed its stronger robustness after several experiments.
3) Then, after replacing the C3 block before the output of the original small target detection head with our designed hybrid representation learning enhancement module, HRLE-C3, the number of FLOPs and parameters slightly decreased compared to ablation 2, from 2.6 to 2.5 G and 1.10 M to 1.09 M, respectively. However, the mAP still increased by 0.1%. The experimental

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results demonstrate that our designed hybrid learning enhancement module, HRLE-C3, fully combines the advantages of convolution and self-attention, which can maintain good performance while reducing the number of FLOPs and parameters and shows its effectiveness in detecting small targets.

4) Finally, after using our PBCE loss designed for the SAR image target detection dataset, the mAP increased from 98.1% to 98.3%. At the same time, the number of FLOPs and parameters remained the same as in ablation 3, which proves that our loss PBCE designed for detection training of SAR image target detection datasets is effective. Moreover, it adaptively adjusts to the SAR image datasets and the target detection task to adjust to a suitable state, improving the model's performance.

In summary, the final HRLE-SARDet that we designed is comparable with YOLOV5n, although the FLOPs decreased by approximately 40.4%, the number of parameters decreased by approximately 38.1%, and the mAP increased by 0.7%, which proves that our designed HRLE-SARDet model is more lightweight while achieving better performance. The visualization comparison results of the ablation experiments are shown in Fig. 10.

2) Comparison Results With the Traditional CFAR Methods on the SSDD Dataset: To further verify the effectiveness of our method, we compare HRLE-SARDet with traditional detection algorithms, such as CFAR and its improved algorithms. Because the statistical indicators of the traditional methods and experimental process and the in-depth learning methods are mostly inconsistent, we only compared the performance of CA-CFAR [50] and OS-CFAR [51] on the SSDD dataset, as shown in Table IV. CA-CFAR is an algorithm to determine whether a pixel belongs to a target by estimating the local environment and time-dependent noise level in the reference window according to the set threshold. OS-CFAR needs high computing power to detect multiple targets. The experimental results in Table IV show that the accuracy and $F1$ score of these traditional methods are far lower than those of our methods. At the same time, compared with traditional methods, our method based on deep learning does not need to manually design the feature extraction process and threshold setting and has strong generalization performance. If we train a new SAR image target detection dataset, our model is more convenient to detect and recognize from end-to-end than the traditional method. Its performance is not inferior to the traditional method, and the model is easier to design and deploy. However, traditional methods also have unique advantages. Combining traditional SAR target recognition and detection methods with deep learning methods is also a very valuable research direction for the future.

3) Comparison Results With the Latest Target Detection Methods on the SSDD Dataset: To continue validating the powerful detectability of our proposed HRLE-SARDet model, we experimented and compared some commonly used and the latest single- and two-stage target detection methods on optical images, such as YOLOV3, YOLOV5, YOLOX, FPN, Faster R-CNN, Cascade R-CNN, CenterNet, and other methods. All parameter settings in the comparison experiments were roughly kept the same for each method, and the datasets and proportions were identical to ensure the fairness of the comparison experiments. The experimental results are shown in Table V. Our proposed method achieves the lowest computational and parametric quantities while exceeding the accuracy of these comparison methods and the highest comprehensive performance of the model. We only have a 0.4% drop in mAP with only approximately one-twentieth the number of parameters and FLOPs of YOLOV5. The predicted visualization of each detection method is shown in Fig. 11. Our proposed method has better detection performance for small targets and fewer missed false detections than other target detection methods.

TABLE IV
COMPARISON WITH THE LATEST TRADITIONAL CFAR METHODS ON THE SSDD DATASET

Method	P(%)	R(%)	F1(%)
CA-CFAR [50]	85.9	98.1	91.6
OS-CFAR [51]	84.2	98.5	90.2
HRLE-SARDet	95.4	95.4	95.4

TABLE V
COMPARISON WITH THE LATEST TARGET DETECTION METHODS ON THE SSDD DATASET

Method	Params(M)	FLOPs(G)	mAP(%)
YOLOX-Nano [34]	0.90	2.55	90.7
CenterNet [52]	14.21	20.4	95.1
YOLOV3-tiny [33]	8.9	12.9	95.8
YOLOV3 [33]	61.52	77.54	96.2
Faster R-CNN [53]	41.12	91.41	96.4
FPN [45]	63.56	71.65	96.5
Cascade R-CNN [54]	69.17	119.05	96.8
YOLOX-s [34]	8.94	26.64	97.2
YOLOV5m [35]	20.85	47.9	98.8
HRLE-SARDet	1.09	2.5	98.4

4) Comparison Results With the Latest SAR Ship Target Detection Methods on the SSDD Dataset: To prove the superiority of our method, we reviewed various papers of the latest SAR ship target detection methods. Since these methods are complicated, much of the code is not open-source or the code is rather confusing, and we did not directly reproduce them but directly cited the experimental results in their papers and tried to keep a fair comparison. Therefore, choosing the experimental setup and dataset is similar to the paper method comparison. Unfortunately, in the field of ship target detection on SAR images, there are a few detectors designed from the lightweight perspective; so many methods did not publish their two indicators of the number of parameters and FLOPs. Hence, we attempted to choose the best research indicators in the experiment for comparison and compared the results, as shown in Table VI. The results show that our HRLE-SARDet can

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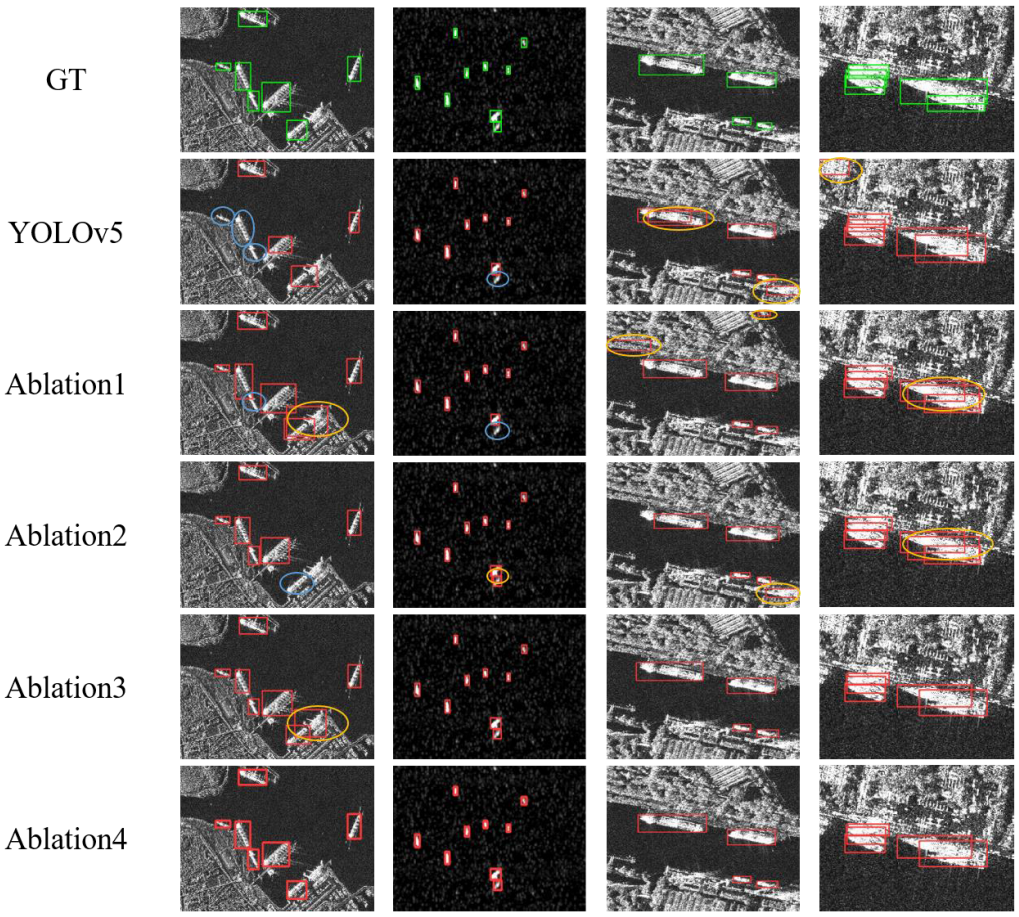

Fig. 10. Comparison results of ablation experiment visualization on the SSDD dataset. The blue circle indicates the missed target and the yellow circle indicates the target of misdetection.

TABLE VI
COMPARISON WITH THE LATEST SAR SHIP DETECTION METHODS ON THE SSDD DATASET

Method	Params(M)	FLOPs(G)	mAP(%)
FBR-Net [30]	32.5	-	94.1
CenterNet++ [55]	-	-	95.1
LPEDet [22]	5.68	18.38	97.4
AFSar [20]	-	9.86	97.7
SII-Net [56]	-	-	95.5
MSSDNet [57]	-	-	95.6
HRLE-SARDet	1.09	2.5	98.4

outperform the latest SAR ship target detectors by far with SOTA with approximately 1 M parameters and 2.5 G FLOPs.

5) Comparison Results in Inshore and Offshore Conditions on the SSDD Dataset: To test and verify the performance of our method under inshore and offshore conditions, we added several groups of experiments to the SSDD test dataset. We test the trained models in the training set on the divided inshore and offshore test sets and use a variety of different methods for comparison. To ensure the fairness of the experiment, our experimental settings are basically consistent with the previous description. The experimental results are shown in Table VII. According to the experimental results in the table, it can be seen that our model maintains extremely high detection performance in both inshore and offshore conditions. Especially in the inshore scene where detection is difficult, our model has achieved the highest detection performance, and the parameters and FLOPs are also kept at extremely low values.

D. Comparison Results With the Latest Target Detection Methods on the AIR-SARShip-2.0 Dataset

In addition to SSDD, a small SAR image ship target detection dataset, we also conducted relevant experiments on another small SAR image ship, the AIR-SARShip-2.0 dataset, to verify the superiority of our model. The experimental results are shown in Table VIII. We mainly compared several lightweight small models of the YOLO series. According to the experimental results in Table VIII, we can see that our HRLE-SARDet model not only achieved the highest accuracy but also the lowest number of parameters and FLOPs. Although the number of parameters is slightly higher than that of YOLOX-Nano, the mAP has increased by more than 12 percentage points. The visualization detection results of our model are shown in Fig. 12.

E. Comparison Results on the HRSID Dataset and Latest SAR Ship Target Detection Methods

To verify the generalizability and robustness of our proposed HRLE-SARDet, we also validate its performance on

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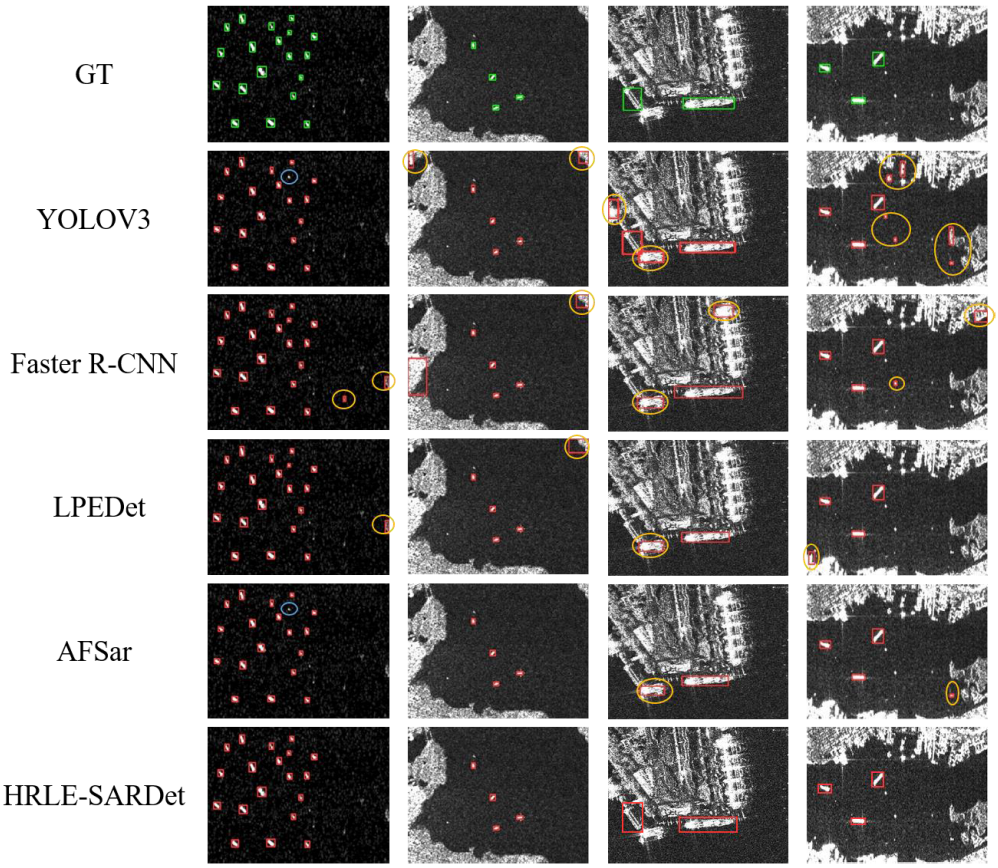

Fig. 11. Comparison of the visualization detection results with the latest method. The blue circle indicates the missed target and the yellow circle indicates the target of misdetection.

TABLE VII
COMPARISON OF INSHORE AND OFFSHORE CONDITIONS ON THE SSDD DATASET

Method	Params(M)	FLOPs(G)	AP(%)		mAP(%)
			inshore	offshore	
YOLOV3-tiny [33]	8.9	12.9	86.3	99.6	95.79
YOLOV4-tiny [58]	5.92	16.1	87.0	99.0	94.8
YOLOX-Nano [34]	0.90	2.55	90.2	90.8	90.7
YOLOV5n [35]	1.76	4.2	92.8	99.3	97.7
HRLE-SARDet	1.09	2.5	93.1	99.1	98.4

TABLE VIII
COMPARISON WITH THE LATEST TARGET DETECTION METHODS ON THE AIR-SARSHIP-2.0 DATASET

Method	Params(M)	FLOPs(G)	mAP(%)
YOLOV3-tiny [33]	8.9	33.0	76.51
YOLOV4-tiny [58]	5.92	98.9	79.12
YOLOX-Nano [34]	0.90	6.52	66.9
YOLOV5n [35]	1.76	4.2	79.0
HRLE-SARDet	1.09	2.5	79.2

the HRSID dataset. HRSID is a large-scale publicly available SAR ship target detection dataset that is many times larger than the SSDD and AIR-SARShip-2.0 Dataset data volume, which further tests the feature extractability of our lightweight model. In addition, we compared the results with the latest SAR ship target detection method using the HRSID dataset, and the comparison results are shown in Table IX. The comparison results show that when training this large SAR ship target detection dataset, despite using the fewest parameters and FLOPs, our HRLE-SARDet model maintains the highest accuracy except YOLOV5, but the number of parameters and FLOPs of HRLE-SARDet is only approximately one-twentieth of YOLOV5. The effectiveness of the HRLE-SARDet method and the strong generalizability and robustness of the model are demonstrated. The visualization detection results of our model are shown in Fig. 13.

F. Comparison Results on the MSAR-1.0 Dataset and Latest SAR Ship Target Detection Methods

To verify that our proposed model has good performance in detecting single-class SAR targets and strong detection and

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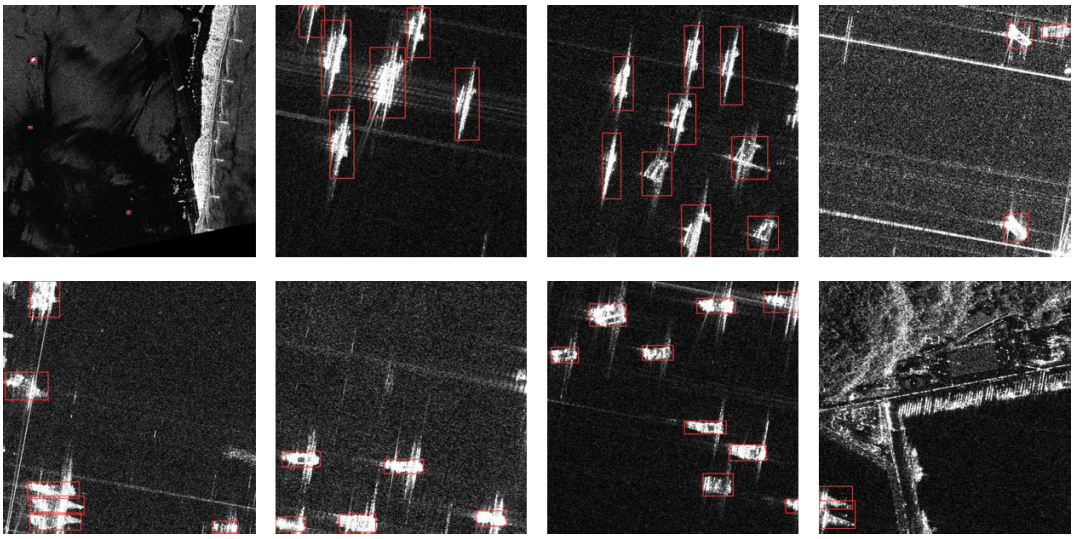

Fig. 12. Visualization detection results of HRLE-SARDet on the AIR-SARShip-2.0 dataset.

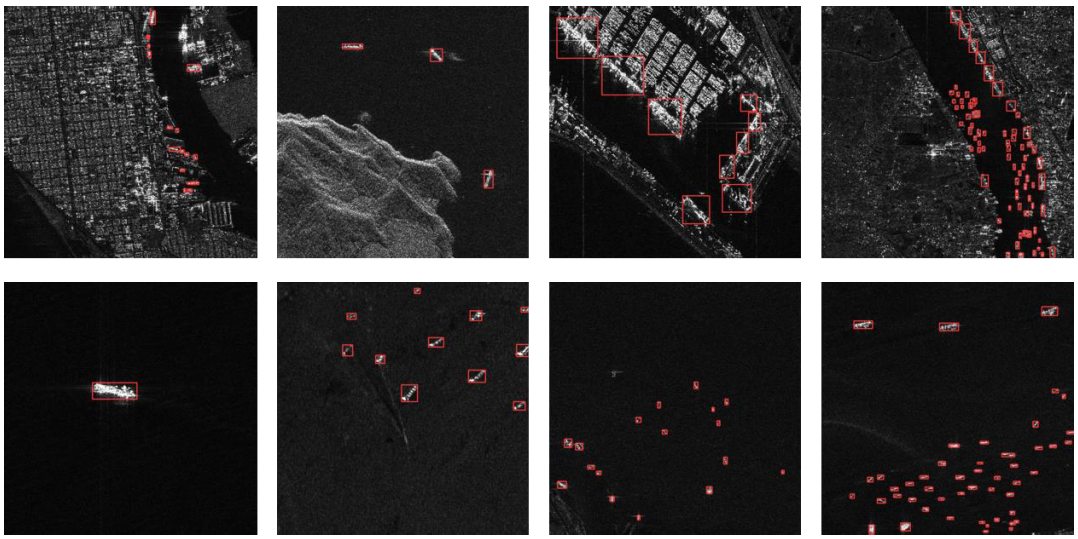

Fig. 13. Visualization detection results of HRLE-SARDet on the HRSID dataset.

TABLE IX
COMPARISON WITH THE LATEST TARGET DETECTION METHODS ON THE HRSID DATASET

Method	Params(M)	FLOPs(G)	mAP(%)
SAR-net [59]	42.6	104.2	84.7
RetinaNet [60]	36.3	127.91	84.8
YOLOV3 [33]	61.52	121.15	85.0
YOLOX [34]	26.64	8.94	87.2
CenterNet2 [61]	71.6	-	89.5
LPEDet [22]	5.68	18.38	89.7
YOLOV5 [35]	20.85	47.9	93.5
HRLE-SARDet	1.09	2.5	92.5

recognizability in detecting multiclass SAR targets, we conducted many validations and comparison experiments on a recently released large-scale multiclass SAR target detection dataset, i.e., MSAR-1.0 [21], [49]. As seen from the results in Table X, our proposed HRLE-SARDet model achieves competitive performance with a minimal number of parameters and FLOPs. We also conducted comparison tests on this dataset using other models of the YOLO series, such as YOLOV3 [33] and YOLOV4-tiny. The experimental setup remained almost identical, and our model improved the mAP by approximately 17% points compared to YOLOV3 with one-sixtieth of the number of parameters and FLOPs. Compared to YOLOV-4tiny, our model improves the mAP by more than 19% points with approximately one-sixtieth of its number of parameters and approximately one-seventh of its FLOPs. The number of parameters and FLOPs of a detection model has a disastrous effect on the detection performance at a low level because a certain number of parameters and FLOPs are necessary to support the learning to obtain better detection accuracy. However, our proposed HRLE-SARDet model achieves a high mAP with both the number of parameters and FLOPs at the minimum, which indicates that the HRLE-SARDet model has

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TABLE X
COMPARISON WITH THE LATEST TARGET DETECTION METHODS ON THE MSAR-1.0 DATASET

Method	Params(M)	FLOPs(G)	AP(%)				mAP(%)
			ship	oil-tank	airplane	bridge	
YOLOV3 [33]	61.52	154.6	91.2	79.9	82.3	32.2	71.4
YOLOV4-tiny [58]	5.92	16.1	93.5	69.7	48.7	64.5	69.1
YOLOX-m [34]	12.45	65.28	96.3	91.4	82.2	79.6	87.4
HRLE-SARDet	1.09	2.5	96.2	85.9	85.4	86.0	88.4

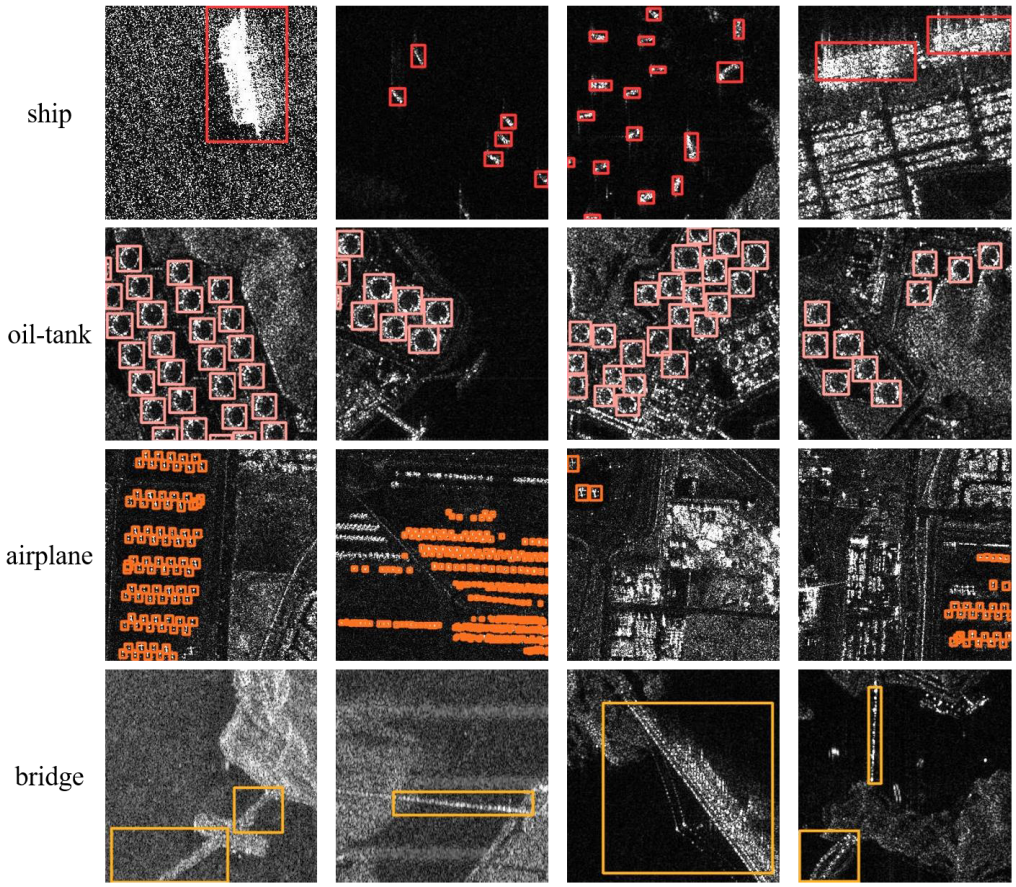

Fig. 14. Visualization detection results of HRLE-SARDet on the MSAR-1.0 dataset.

strong target detectability. The visualization detection results of our model on the MSAR-1.0 dataset are shown in Fig. 14.

G. Partial Visualization of the Scattering Feature Map

The basic purpose of SAR imaging is to obtain a high-resolution 2-D image of the target and obtain the position and shape information of the target. In fact, the geometric structure, physical parameters, materials, and other information of the target contained in the SAR image can affect the electromagnetic response characteristics of the target relative to the radar signal from different aspects. Therefore, SAR images are quite different from optical images in terms of imaging mechanism, geometric features, and radiation features. Electromagnetic characteristics generally include scattering center characteristics and polarization characteristics. In the optical region, the electromagnetic scattering coherent combination of multiple local scattering sources forms the echo of radar targets, and these scattering sources are also common scattering centers. The scattering center feature of the target generally represents the geometric feature of the target, while the polarization feature generally reflects the surface roughness information of the target.

The target, position, structure, and surface morphology of various ground objects are often different, so the response to radar beams (electromagnetic beams) is also different. Due to various factors, such as radar, polarization mode, incident position, and incident angle of the waveband, the hue and texture of the ground object on the image will also be different. The backscattering echo intensity of ground objects is represented by the gray level of the radar image, and the

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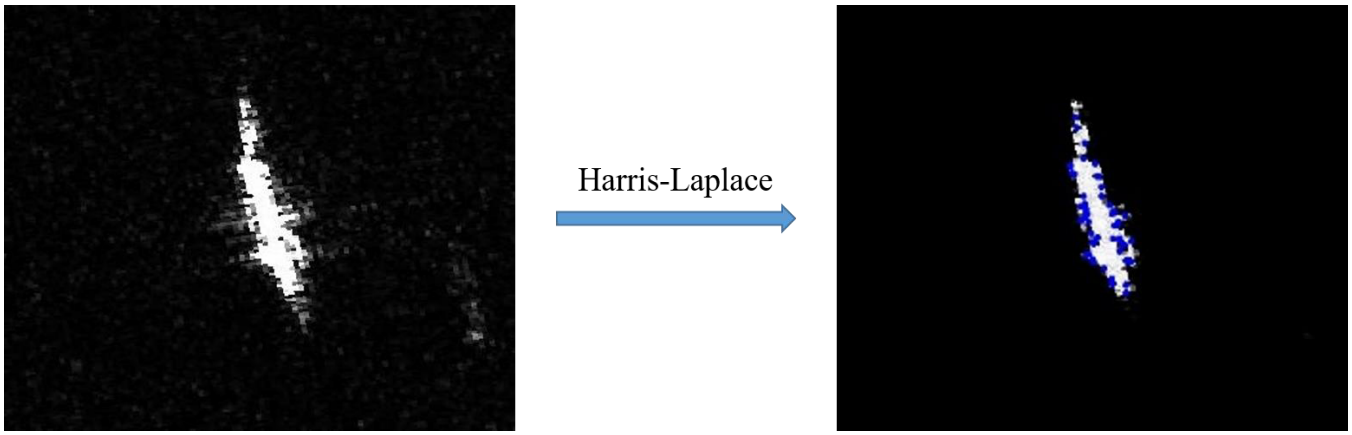

Fig. 15. Schematic of strong scattering point extraction of the ship target in the SAR image.

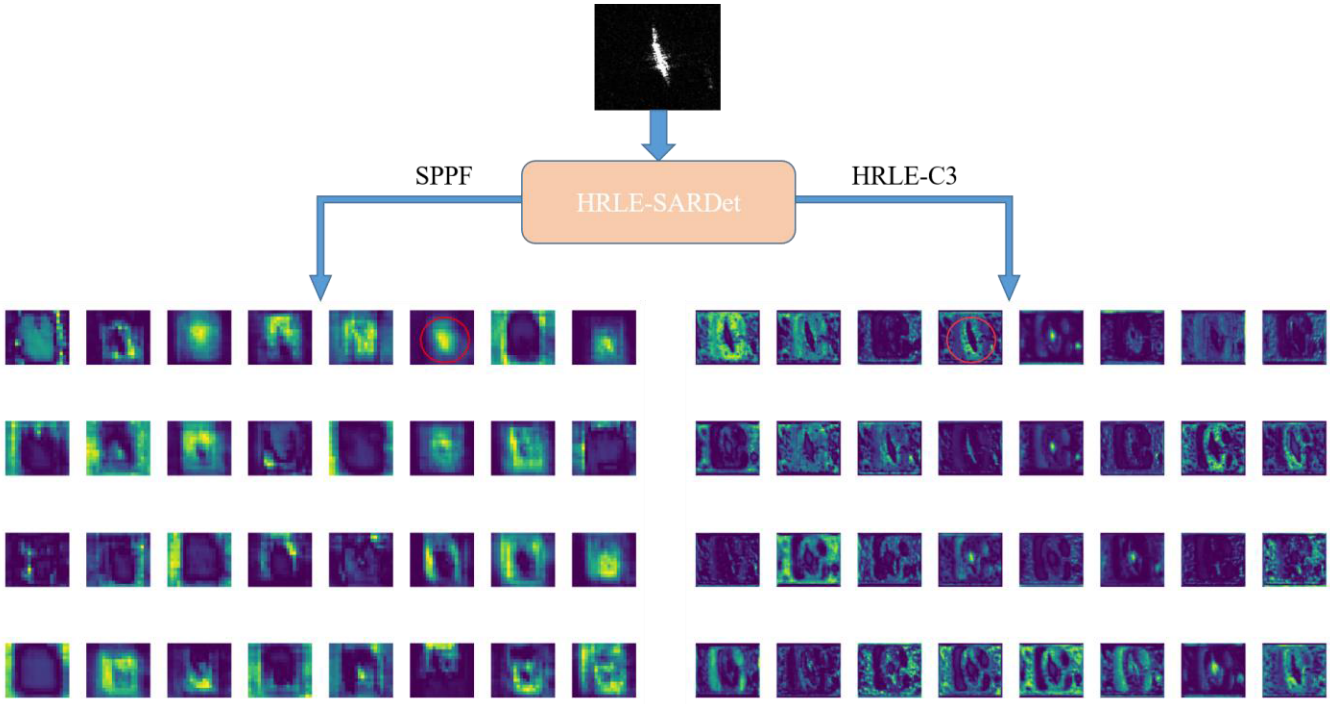

Fig. 16. Partial channel feature maps of SPPF and HRLE-C3. The red circle indicates the scattering geometry information.

backscattering cross section and backscattering coefficient of the radar are usually used to express the backscattering.

The image information formed by the backscattering of the ground object targets constitutes the main information in the SAR image and is also the reflection of the ground object targets to the radar beam. Target scattering feature information is usually the basic information that we use to understand and apply SAR images. These features are mainly composed of geometric features, texture features, polarization features, noise features, and gray-level statistical features of the target.

Taking ships as an example, the basic structure of ships is fundamentally the same and is mainly composed of five parts: bow, hull, cabin, deck, and power plant. Because ships are mainly made of metal materials and SAR has good detection performance for metal materials, ships in SAR images contain rich scattering information. The scattering mechanism and degree of each part of the ship are different, and the final appearance is also different.

In the high-resolution SAR image, the scattering information of the target in the SAR image is mainly composed of the scattering information of each target component. Due to the different scattering mechanisms of target components, the scattering information used for target detection and recognition in SAR images mainly includes target geometric features, textural features, polarization features, and gray-level statistical features.

Because the geometric shape of the target is a unique mark in SAR images, it is effective to use geometric features for target detection and recognition in SAR images. From the previous introduction and SAR images, we can see that the general target is represented in the form of discrete strong scattering points. Through the detection and discrimination of

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such obvious point targets, the targets in SAR images can be recognized and located. In addition, due to the large scattering difference between the whole target or its subcomponents and the background in the SAR image, the boundary points of the target are often formed at the abrupt change in gray level, so the target in the SAR image easily forms a boundary linear target with strong edge information.

The focus of this article is to detect typical targets, such as ships in SAR images. Usually, these typical targets show certain scattering characteristics in the image and are mainly composed of a collection of multiple strong scattering points. We extract their strong scattering point information by using the Harris–Laplace corner detector [62], as shown in Fig. 15.

The combination of these strong scattering points constitutes the main target information. By observing the distribution of scattering points, we find that they are generally distributed at the edge of the target, so they reflect the geometric characteristics of the target to a certain extent.

Because convolution and self-attention modules are usually designed by different design theories, convolution and self-attention are generally considered two different methods, but they are both powerful representational learning methods in the field of deep learning. Traditional convolution uses an aggregation function on the local receptive field. In image processing, the inherent characteristics bring important inductive bias. In contrast to convolution, the self-attention module can dynamically calculate the attention weight through a weighted average operation based on the input feature context and similarity function between relevant pixel pairs. Therefore, the self-attention module can flexibly and adaptively focus on the edge region to capture more geometric features. Therefore, this method, which combines the advantages of convolution and self-attention, can pay more attention to the geometric feature information brought by this strong scattering point and is very suitable for application in SAR image target detection. Finally, through a large number of experiments, to verify the design, we debug our model to the optimal state to achieve the optimal SAR target detection performance.

In addition, we visualized the partial channel feature maps of the SPPF layer at the end of the backbone of the model and the HRLE-C3 layer in front of the output detection head of the small target in the SAR image. The channel feature maps are shown in Fig. 16. It can be seen in the figure that the red circle indicates the scattering geometry information. Therefore, our model backbone, LSFEBackbone, and hybrid representation learning enhancement module, HRLE-C3, can extract the scattering geometry information of the target very well.

V. Conclusion

This article introduces a lightweight SAR target detection algorithm called HRLE-SARDet based on hybrid representation learning enhancement, which proposes an SAR image target detection algorithm. The approach is suitable for deployment to embedded devices from a lightweight perspective and combines it with the unique feature information characteristics of SAR images.

1) We introduce YOLOv5 as our lightweight benchmark model, based on which we design a scattering feature extraction backbone, LSFEBackbone. LSFEBackbone is more adapted to the characteristics of SAR image datasets and achieves a certain balance in speed and accuracy.
2) The multiscale neck, BiFPF, is redesigned for the multiscale characteristics of SAR targets, which more fully integrates the deep and shallow semantic feature information of different scales to improve the detection effect of targets of different scales.
3) To better extract the scattering information of small targets and improve the detection accuracy of small targets, we design a hybrid representation learning enhancement module called HRLE-C3, which combines the respective advantages of CNN and self-attention to complement each other and can improve the detection performance of SAR small targets while reducing the number of parameters and FLOPs.
4) A more suitable and flexible loss function for the target detection task of SAR image datasets is redesigned, PBCE, based on the BCE loss function. PBCE allows the loss function to adapt to the target task and dataset easily. Finally, extensive experimental results on three public benchmark SAR ship target detection datasets and a newly released large multiclass target SAR dataset, MSAR-1.0, show that our HRLE-SARDet can achieve mAP values of 98.4%, 79.2%, 92.5%, and 88.4% on the SSDD, AIR-SARShip-2.0 Dataset, HRSID, and MSAR-1.0 datasets, respectively, with only 1.09 M and 2.5 G of parameters and FLOPs, which is an excellent performance.

References

[1] M. Shahzad, M. Maurer, F. Fraundorfer, Y. Wang, and X. X. Zhu, "Buildings detection in VHR SAR images using fully convolution neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1100–1116, Feb. 2019.

[2] L. Du, H. Dai, Y. Wang, W. Xie, and Z. Wang, "Target discrimination based on weakly supervised learning for high-resolution SAR images in complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 461–472, Jan. 2020.

[3] G. Yang, H.-C. Li, W. Yang, K. Fu, Y.-J. Sun, and W. J. Emery, "Unsupervised change detection of SAR images based on variational multivariate Gaussian mixture model and Shannon entropy," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 826–830, May 2019.

[4] P. Iervolino and R. Guida, "A novel ship detector based on the generalized-likelihood ratio test for SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3616–3630, Aug. 2017.

[5] C. H. Gierull and I. Sikaneta, "A compound-plus-noise model for improved vessel detection in non-Gaussian SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1444–1453, Mar. 2017.

[6] C. P. Schwegmann, W. Kleynhans, and B. P. Salmon, "Manifold adaptation for constant false alarm rate ship detection in South African oceans," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3329–3337, Jul. 2015.

[7] L. M. Kaplan, "Improved SAR target detection via extended fractal features," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 436–451, Apr. 2001.

[8] M. Tello, C. Lopez-Martinez, and J. J. Mallorqui, "A novel algorithm for ship detection in SAR imagery based on the wavelet transform," *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 2, pp. 201–205, Apr. 2005.

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[9] S. Song and J. Yang, "Ship detection in polarimetric SAR images via tensor robust principle component analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 3152–3155.

[10] S. Song, B. Xu, Z. Li, and J. Yang, "Ship detection in SAR imagery via variational Bayesian inference," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 319–323, Mar. 2016.

[11] X. Leng, K. Ji, K. Yang, and H. Zou, "A bilateral CFAR algorithm for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1536–1540, Jul. 2015.

[12] Y. Liu, M.-H. Zhang, P. Xu, and Z.-W. Guo, "SAR ship detection using sea-land segmentation-based convolutional neural network," in *Proc. Int. Workshop Remote Sens. Intell. Process. (RSIP)*, May 2017, pp. 1–4.

[13] M. Kang, K. Ji, X. Leng, and Z. Lin, "Contextual region-based convolutional neural network with multilayer fusion for SAR ship detection," *Remote Sens.*, vol. 9, no. 8, p. 860, 2017.

[14] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2019.

[15] L. Yu, H. Wu, Z. Zhong, L. Zheng, Q. Deng, and H. Hu, "TWC-Net: A SAR ship detection using two-way convolution and multiscale feature mapping," *Remote Sens.*, vol. 13, no. 13, p. 2558, Jun. 2021.

[16] Z. Sun et al., "An anchor-free detection method for ship targets in high-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7799–7816, 2021.

[17] X. Ma, K. Ji, B. Xiong, L. Zhang, S. Feng, and G. Kuang, "Light-YOLOv4: An edge-device oriented target detection method for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10808–10820, 2021.

[18] Z. Sun, X. Leng, Y. Lei, B. Xiong, K. Ji, and G. Kuang, "BiFA-YOLO: A novel YOLO-based method for arbitrary-oriented ship detection in high-resolution SAR images," *Remote Sens.*, vol. 13, no. 21, p. 4209, Oct. 2021.

[19] S. Chen, J. Zhang, and R. Zhan, "R^{2}FA-Det: Delving into high-quality rotatable boxes for ship detection in SAR images," *Remote Sens.*, vol. 12, no. 12, p. 2031, Jun. 2020.

[20] H. Wan et al., "AFSar: An anchor-free SAR target detection algorithm based on multiscale enhancement representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5219514, doi: 10.1109/TGRS.2021.3137817.

[21] R. Xia et al., "CRTransSar: A visual transformer based on contextual joint representation learning for SAR ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, Mar. 2022.

[22] Y. Feng et al., "A lightweight position-enhanced anchor-free algorithm for SAR ship detection," *Remote Sens.*, vol. 14, no. 8, p. 1908, Apr. 2022, doi: 10.3390/rs14081908.

[23] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[24] S. Xian, W. Zhirui, S. Yuanrui, D. Wenhui, Z. Yue, and F. Kun, "AIR-SARShip-1.0: High-resolution SAR ship detection dataset," *J. Radars*, vol. 8, no. 6, pp. 852–863, 2019.

[25] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[26] Y. Wang and H. Liu, "A hierarchical ship detection scheme for high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 4173–4184, Oct. 2012.

[27] W. Huo, Y. Huang, J. Pei, Q. Zhang, Q. Gu, and J. Yang, "Ship detection from ocean SAR image based on local contrast variance weighted information entropy," *Sensors*, vol. 18, no. 4, p. 1196, 2018.

[28] Z. Shi, X. Yu, Z. Jiang, and B. Li, "Ship detection in high-resolution optical imagery based on anomaly detector and local shape feature," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4511–4523, Aug. 2014.

[29] N. Liu, Z. Cao, Z. Cui, Y. Pi, and S. Dang, "Multi-scale proposal generation for ship detection in SAR images," *Remote Sens.*, vol. 11, no. 5, p. 526, Mar. 2019.

[30] J. Fu, X. Sun, Z. Wang, and K. Fu, "An anchor-free method based on feature balancing and refinement network for multiscale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1331–1344, Feb. 2021.

[31] Y.-L. Chang, A. Anagaw, L. Chang, Y. Wang, C.-Y. Hsiao, and W.-H. Lee, "Ship detection based on YOLOv2 for SAR imagery," *Remote Sens.*, vol. 11, no. 7, p. 786, Apr. 2019.

[32] Z. Zhou et al., "FSODS: A lightweight metalearning method for few-shot object detection on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5232217, doi: 10.1109/TGRS.2022.3192996.

[33] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[34] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[35] Ultralytics. *YOLOv5*. Accessed: 2021. [Online]. Available: https://github.com/ultralytics/yolov5

[36] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[37] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[39] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-ResNet and the impact of residual connections on learning," in *Proc. AAAI Conf. Artif. Intell.*, 2016, pp. 1–7.

[40] H. Jie, S. Li, S. Gang, and S. Albanie, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 1–12, Aug. 2017.

[41] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.

[42] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.

[43] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[44] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[45] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[46] K. Han, A. Xiao, E. Wu, J. Guo, C. Xu, and Y. Wang, "Transformer in transformer," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 15908–15919.

[47] X. Pan et al., "On the integration of self-attention and convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 815–825.

[48] Z. Leng et al., "PolyLoss: A polynomial expansion perspective of classification loss functions," 2022, *arXiv:2204.12511*.

[49] J. Chen et al. (2022). *Large-Scale Multi-Class SAR Image Target Detection Dataset-1.0[OL]*. Journal of Radars. [Online]. Available: https://radars.ac.cn/web/data/getData?dataType=MSAR

[50] C. Kuang, C. Wang, B. Wen, Y. Hou, and Y. Lai, "An improved CA-CFAR method for ship target detection in strong clutter using UHF radar," *IEEE Signal Process. Lett.*, vol. 27, pp. 1445–1449, 2020.

[51] E. Hyun and J.-H. Lee, "A new OS-CFAR detector design," in *Proc. 1st ACIS/JNU Int. Conf. Comput., Netw., Syst. Ind. Eng.*, May 2011, pp. 133–136.

[52] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 6569–6578.

[53] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[54] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 6154–6162.

[55] H. Guo, X. Yang, N. Wang, and X. Gao, "A CenterNet++ model for ship detection in SAR images," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107787.

[56] N. Su, J. He, Y. Yan, C. Zhao, and X. Xing, "SII-Net: Spatial information integration network for small target detection in SAR images," *Remote Sens.*, vol. 14, no. 3, p. 442, Jan. 2022, doi: 10.3390/rs14030442.

[57] K. Zhou, M. Zhang, H. Wang, and J. Tan, "Ship detection in SAR images based on multi-scale feature extraction and adaptive feature fusion," *Remote Sens.*, vol. 14, no. 3, p. 755, Feb. 2022, doi: 10.3390/rs14030755.

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[58] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "Scaled-YOLOv4: Scaling cross stage partial network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13029–13038.
[59] S. Gao, J. M. Liu, Y. H. Miao, and Z. J. He, "A high-effective implementation of ship detector for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3115121.
[60] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.
[61] X. Zhou, V. Koltun, and P. Krähenbühl, "Probabilistic two-stage detection," 2021, *arXiv:2103.07461*.
[62] E. Rosten and T. Drummond, "Machine learning for high-speed corner detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2006, pp. 430–443.

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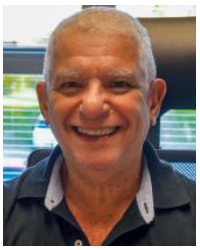

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